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## TIMESCALEDB PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Databases

TOTAL: 10 QUESTIONS

**Q1** What type of database is TimescaleDB and what are its core strengths?

Threat Model

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**Q6** How do you monitor and tune slow queries in TimescaleDB?

Observability

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**Q2** How does TimescaleDB guarantee consistency and handle transactions?

Architecture

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**Q7** What backup and restore strategies does TimescaleDB support?

Lifecycle

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**Q3** What index types does TimescaleDB support and when do you use each?

Configuration

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**Q8** How do you handle schema migrations in TimescaleDB?

Compliance

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**Q4** How do you design a schema or data model in TimescaleDB?

Hardening

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**Q9** What security features does TimescaleDB provide?

Testing

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**Q5** How does TimescaleDB scale horizontally?

SSO / IdP

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**Q10** What are common anti-patterns when using TimescaleDB in production?

Tuning

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## ANSWER KEY &amp; EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls &amp; reference

VERIFIED SOLUTIONS

**A1 TimescaleDB is a relational, document, key-value,** Mitigation

TimescaleDB is a relational, document, key-value, graph, or time-series store. Its strengths such as ACID guarantees, horizontal scale, or flexible schema guide the workloads it is best suited for.

**A6 TimescaleDB provides a slow query log,** Observability

TimescaleDB provides a slow query log, explain/explain plan, or profiling tools to surface inefficient queries. Common fixes include adding indexes, rewriting queries, or increasing cache size.

**A2 TimescaleDB provides either full ACID** Primitives

TimescaleDB provides either full ACID transactions or eventual consistency depending on its CAP theorem positioning. Transaction support varies from none (simple K/V stores) to serializable isolation (relational DBs).

**A7 TimescaleDB supports logical backups (export** Automation

TimescaleDB supports logical backups (export SQL or BSON), physical backups (filesystem snapshot), and continuous replication to a standby. Point-in-time recovery requires WAL/oplog archiving on top of backups.

**A3 TimescaleDB offers B-tree, hash, full-text, spatial,** Hardening

TimescaleDB offers B-tree, hash, full-text, spatial, or composite indexes depending on the engine. Choose based on query patterns: B-tree for range queries, hash for equality lookups, full-text for search.

**A8 Schema migrations in TimescaleDB are** Governance

Schema migrations in TimescaleDB are managed with versioned migration files applied in order by a migration tool. Migrations should be idempotent and tested in staging before production to avoid downtime.

**A4 Schema design in TimescaleDB involves** Gotchas

Schema design in TimescaleDB involves normalizing relations (relational DB) or denormalizing for access patterns (document/key-value). Understanding query needs upfront prevents costly migrations later.

**A9 TimescaleDB supports role-based access** Assessment

TimescaleDB supports role-based access control, encrypted connections (TLS), encryption at rest, and audit logging. Always restrict user privileges to the minimum needed and disable anonymous access in production.

**A5 TimescaleDB scales via replication (read** Federation

TimescaleDB scales via replication (read replicas) for read-heavy workloads and sharding (partitioning data by key) for write-heavy ones. Some TimescaleDB implementations handle this automatically; others require manual configuration.

**A10 Common mistakes include selecting all** Optimization

Common mistakes include selecting all columns instead of needed fields, missing indexes on frequently queried columns, running migrations without backups, and storing large blobs directly in the database instead of object storage.